Caption generation and selection

Visual region extraction

Linear projection
768 → 512

ROI pooling

Patch features
14 × 14 × 768

Bounding boxes B

CLIP ViT-B/16
Frozen visual encoder

YOLO
Detector



Input image I

V
N × 512

Cascaded semantic fusion

V_o
V_r

Region-linked prompt conditioning

Semantic bank–region alignment

TRAIN captions → noun / verb lemmas

CLIP text encoder

TRAIN noun bank
Fixed embeddings

TRAIN verb bank
Fixed embeddings

Learned semantic aggregation

Learned semantic aggregation

P_n
150 × 512

P_v
150 × 512

V

Object fusion 1

V_1, T_1

Object fusion 2

V_o

V_o

Relation fusion

V_r

V_o, V_r : N × 512

Image-specific prompt–region alignment

YOLO boxes

Image I → UPT
Interaction detector

Region matching
Valid ROI links

Object phrases

Relation phrases

CLIP text encoder
Frozen

t_r t_o Prompt embeddings

V_o

Object prompt adapter

Linked slots

Masked attention

V_o'

Slots: original V_o

V_r

Relation prompt adapter

Linked slots

Masked attention

V_r'

Transformer decoder

Gated fusion + FFN

V_o'
V_r'

Parallel cross-attention

Masked self-attention

×3

Word embedding
Fixed position

Token prefix

Vocabulary projection
+ softmax

Beam search
K = 5

Five candidate captions

y1 y2 y3 y4 y5

Caption reranking

Decoder likelihood D

Image–text alignment C

Object consistency O

Weighted scoring and selection

I + YOLO labels

Final caption

Training: cross-entropy pretraining followed by SCST + cross-entropy.

Frozen parameters

Trainable parameters